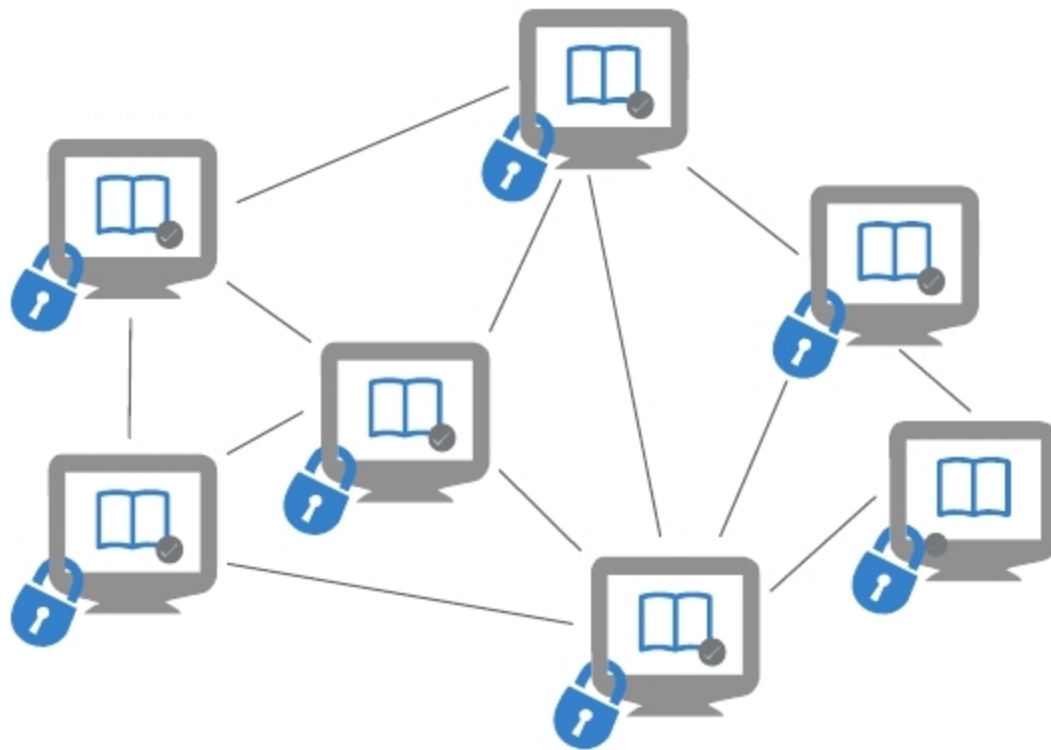


Why the Blockchain is Your Best Bet *for* Cyber Security

Blockchain is a tamper-proof, shared digital ledger that records the history of transactions between the peers in a peer-to-peer network. This article describes how blockchain technology can be used to protect data and the network from cyber-attacks.



Cyber security is a set of technologies, processes and controls designed to protect systems, devices, data, networks and programs from cyber-attacks. It secures data from threats such as theft or misuse, and also safeguards the system from viruses.

In today's world, cyber-attacks are major threats faced by each user. Most of us are responsive to the advertisements on various websites on the Internet, and if asked questions or any personal details, respond without even thinking of the consequences. Sharing one's personal information is very risky, as one may lose whatever one has. In 2016, the search engine Yahoo! faced a major attack and around one billion accounts were compromised. The attackers were able to get the user names, passwords, phone numbers, and security questions and answers of e-mail users. On September 7, 2017, Equifax, one of the largest consumer credit recording

agencies in the world, faced a massive cyber security threat. It is believed that someone gave unauthorised access to the data with this agency, from mid-May to July 2017. Around 145.5 million people felt threatened by the news as they had shared personal information like names, social security numbers, birthdays, addresses and driving license numbers with Equifax.

Many people use weak or default passwords for their personal accounts on some Internet sites, which is very risky as these can be cracked, and their personal details can be compromised. This may be even more risky if people use default passwords for all their sites, just for convenience. If the attackers crack that password, then it can be used for all other sites, and all their personal details, including their credit card and bank details may be harvested. In this digital era, cyber-attacks are a matter of real concern. Cyber criminals are greatly increasing in number and

are attempting to steal financial data, personal identifiable information (PII) as well as identities of the common Internet user.

Businesses, the government, and the private as well as public sectors are continuously fighting against such frauds, malicious bugs and so on. Even as the hackers are increasing their expertise, the ways to cope with their attacks are also improving very fast. One of these ways is the blockchain.

Blockchain: A brief background

Each block in a blockchain contains transaction data, a hash function and hash of the previous block. Blockchain is managed by a peer-to-peer (P2P) network. On the network, no central authority exists and all blocks are distributed among all the users in the network. Everyone in the network is responsible for verifying the data that is shared, and ensuring that no existing blocks are being altered and no false data is being added. Blockchain technology enables direct transaction between two individuals, without the involvement of a third party and hence provides transparency. When a transaction happens, the transaction information is shared amongst everyone in the blockchain network. These transactions are individually time stamped. When these transactions are put together in a block, they are time stamped again as a whole block. Blockchain can be used to prevent cyber-attacks in three ways – by being a trusted system, by being immutable and by network consensus.